NIHAR GUDADHE

Data Scientist & Analyst

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SUMMARY

Data Scientist & Analyst with 2 years 9 months of experience taking problems from raw data to deployed models, across regression, classification, NLP, and anomaly detection. At Influx Control Panel, built and deployed a Random Forest pricing model (FastAPI, Docker) that cut quotation turnaround from days to seconds, and delivered Power BI dashboards behind a 10% lift in sales conversion, a 12% rise in shop-floor throughput, and a 14% drop in product returns. Independently built three end-to-end projects: a complaint triage platform on 150,000 CFPB complaints with fine-tuned RoBERTa, a local RAG question-answering system over 49,877 Amazon reviews, and a shopper purchase-intent model with MLflow tracking and live deployment. Works across Python, SQL, Docker, and CI/CD, and documents assumptions and limitations so results can be trusted.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, PostgreSQL, MySQL, DuckDB, Snowflake, Excel

Machine Learning & Statistics: Regression, Classification, Ensemble Methods (Random Forest, XGBoost, LightGBM), Feature Engineering & Selection, Model Evaluation, Model Calibration, Cross-Validation, Hyperparameter Tuning (GridSearchCV), SMOTE, Anomaly Detection, Hypothesis Testing, Confidence Intervals (Bootstrap), Clustering, Customer Segmentation, Time Series Forecasting, PCA, A/B Testing

Data Analysis & Visualization: Pandas, Power BI, Streamlit, Exploratory Data Analysis, Data Cleaning, Missing Value & Outlier Treatment, NumPy, Matplotlib, Seaborn, DAX, Power BI Service, Tableau

NLP & Generative AI: RoBERTa, Hugging Face Transformers, Tokenization, Word Embeddings (sentence-transformers), Large Language Models (LLMs, Llama 3.2 via Ollama), Prompt Engineering, RAG, LangGraph, LanceDB, LangChain, Named Entity Recognition, Sentiment Analysis, NLTK, spaCy

Deep Learning: PyTorch, TensorFlow, ANN, CNN, RNN, LSTM, GRU, Encoder-Decoder, Attention Mechanism

Computer Vision: Image Classification, Object Detection (YOLO), Image Segmentation, OpenCV

MLOps & Deployment: Scikit-learn, FastAPI, Docker, Git, GitHub, CI/CD (GitHub Actions), MLflow, Evidently AI, Drift Monitoring (PSI), Prefect, Pytest, Render, Jupyter Notebook, VS Code

Cloud & Data Engineering: AWS (S3, IAM), dbt, ETL/ELT Pipelines, Data Warehousing, Snowflake, PySpark, Databricks

Core Concepts: Statistical Inference, Data Modeling, Reproducible Workflows, Data Labeling, Model Cards & Technical Documentation

PROFESSIONAL EXPERIENCE

Data Scientist & Analyst | Influx Control Panel, Nagpur **Feb 2024 - Present**

- **Pricing Prediction Model:** Built a Random Forest regression model in Scikit-learn to estimate costs for complex industrial items, and deployed it behind a real-time FastAPI endpoint running in Docker. Quotes that used to take days now come back in seconds.
- **Data Pipelines & Warehousing:** Set up an automated ETL pipeline that pulls raw CRM leads and billing records into a MySQL staging area, transforms them with dbt, and loads clean, analysis-ready tables into a PostgreSQL warehouse. It replaced the manual, error-prone data pulls the team used to depend on.
- **Sales Analytics:** Designed a 360-degree customer dashboard in Power BI covering account value, engagement, and conversion stage. The sales team used it to prioritize high-value segments, and conversion rates improved by 10%.
- **Production Efficiency:** Built a production line dashboard that tracks cycle times and machine idle time. The bottlenecks it exposed on the shop floor led directly to a 12% increase in throughput.
- **Quality Analysis:** Used anomaly detection to build an automated defect report that links raw material batches to support ticket trends. Engineering used it to catch recurring issues earlier and cut product returns by 14%.

PROJECTS

SQIPE - Enterprise Analytics & ML Pricing Platform | (Internal Project)

Python | Scikit-learn | PostgreSQL | MySQL | dbt | FastAPI | Docker | Power BI

- Wrote automated ETL scripts that pull data from CRM webhooks and Tally transaction logs into one properly structured relational warehouse, bringing together pricing and sales data that used to sit in scattered sources.
- Built feature-engineered pricing models that account for non-linear component costs, switchgear configurations, and historical win-loss ratios, replacing a pricing process that was manual and inconsistent.

- Deployed the model as a FastAPI inference service across multi-container Docker environments, with GitHub Actions handling CI/CD so updates shipped without manual steps.
- Built SQL-backed Power BI dashboards that give leadership live visibility into industrial pipeline risk and throughput.

BankSight: Consumer Complaint Intelligence Platform | (Independent Project)

Python | DuckDB | dbt | Snowflake | AWS S3 | RoBERTa | FastAPI | Docker | Streamlit | Prefect

- Built an end-to-end triage system on 150,000 CFPB complaints that ranks them by their chance of getting relief, so reviewers read the likeliest ones first. The warehouse runs on DuckDB and dbt with automated data tests, and the same dbt models ran on Snowflake with identical monthly KPIs.
- Compared four models on a time-based split, with the selection rule fixed in advance and the test set opened only once. Fine-tuned RoBERTa beat the TF-IDF baseline with a PR-AUC of 0.403 against 0.378 (random is 0.183), and the 95% interval stayed above zero.
- Calibrated the scores (calibration error 0.016) and set a cost-based review threshold that catches 63% of relief cases while flagging 32% of complaints.
- Shipped a FastAPI service (with a Dockerfile) and a Streamlit dashboard with monthly drift monitoring (PSI) on Prefect. Model artifacts sit in private AWS S3 under least-privilege IAM, backed by pytest tests and a model card.

ShopSense: Product Review Q&A System | (Independent Project)

Python | sentence-transformers | LanceDB | Ollama (Llama 3.2 3B) | LangGraph | FastAPI | Streamlit | pytest

- Built a local question-answering system over 49,877 Amazon reviews (50,524 passages) with no paid API. On 19 labelled questions, meaning-based search found a relevant review for all 19, against 16 for keyword search. Used bootstrap intervals to show which differences were clear.
- Audited answers by hand, Found and fixed a bug in the citation checker, removed 2 wrong “no answer” replies, and added code that deletes uncited sentences and checks quotes against the cited review. Reported the rest honestly: 8 of 10 answers had every claim in the reviews, but only about 2 had every citation number right.
- Built a LangGraph agent that sends number questions to read-only pandas tools and opinion questions to review search. It routed 14 of 18 rephrased questions correctly and refused all 17 unanswerable ones. Served it through FastAPI and Streamlit.
- Tested prompt injection with 15 poisoned reviews, and the model obeyed none. Cut wrongly flagged honest reviews from 74 to 2 out of 50,524. Wrote 48 automated tests.

Shoppers Intent MLOps | (Independent Project)

Python | FastAPI | MySQL | PostgreSQL | Scikit-learn | XGBoost | LightGBM | MLflow | Evidently AI | Docker | Render

- Trained and compared Random Forest, XGBoost, and LightGBM models to predict whether an online shopper will complete a purchase, tuning each with GridSearchCV. The best model reached 91.8% accuracy and 97.3% ROC-AUC across 122,330 sessions.
- The data was heavily skewed toward non-purchases, so SMOTE was applied along with feature scaling and selection to fix the imbalance. Recall rose to 94.3%, which means far fewer real buyers were missed.
- Used MLflow to track every experiment and version each model, then wrapped the final model in a FastAPI service containerized with Docker. GitHub Actions ran automated testing and deployment on every push.
- Added drift monitoring with Evidently AI and a daily auto-retraining job so the model does not go stale, and deployed the whole system live on Render instead of leaving it in a notebook.

EDUCATION

B.E. in Computer Science Engineering **2012 - 2016**

Sipna College of Engineering, Amravati

CERTIFICATIONS

- Diploma in Machine Learning using Python - Alison
- Data Science Essentials with Python - Cisco Networking Academy